Unit: Unit 8: Quadratics **Name:** ____________
Topic: Definition of Quadratics and Standard Form **Date:** __________

Foundation (Jalapeno)

Q1. What is the standard form of a quadratic function?
(A) $ax^2 + bx + c$
(B) $ax^2 - bx + c$
(C) $ax^2 + bx - c$
(D) $ax^2 - bx - c$
(E) $ax + bx + c$

Q2. A spaceship is traveling through space with a height given by $h(t) = -t^2 + 4t + 1$. What type of function is $h(t)$?
(A) Linear
(B) Quadratic
(C) Exponential
(D) Polynomial
(E) Rational

Q3. Which of the following is a quadratic function in standard form?
(A) $x^2 - 3x + 2$
(B) $x^2 + 2x - 3$
(C) $x + 2$
(D) $x^2 - 2$
(E) $x - 2$

Q4. A scientist is studying the trajectory of a planet with a path given by $p(x) = 2x^2 + 5x - 1$. What is the coefficient of the x^2 term?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

Q5. What is the key feature of a quadratic function that determines its direction?
(A) Vertex
(B) Axis of symmetry
(C) Leading coefficient
(D) Constant term
(E) Coefficient of the x term

Q6. A lab experiment involves a quadratic relationship between pressure P and volume V given by $P(V) = 3V^2 - 2V + 1$. What type of relationship is this?
(A) Direct variation
(B) Inverse variation
(C) Quadratic relationship
(D) Exponential relationship
(E) Linear relationship

Q7. Which of the following quadratic functions has a negative leading coefficient?
(A) $x^2 + 2x + 1$
(B) $-x^2 + 2x + 1$
(C) $2x^2 + 2x + 1$
(D) $x^2 - 2x + 1$
(E) $-2x^2 + 2x + 1$

Q8. A physics problem involves a quadratic function $d(t) = t^2 - 4t + 4$ representing the distance of an object over time. What is the vertex of this function?
(A) (0, 0)
(B) (2, 0)
(C) (2, -4)
(D) (0, 4)
(E) (4, 0)

Open Ended Questions (FRQ)

Q9. Describe the key features of the quadratic function $f(x) = x^2 + 2x - 3$. Be sure to include the vertex, axis of symmetry, and leading coefficient.

Q10. Write a quadratic function in standard form to represent the height of a projectile launched from the ground with an initial velocity of 20 m/s. Assume the acceleration due to gravity is -9.8 m/s^2.

Chef's Tips

- Emphasize the importance of identifying the leading coefficient to determine the direction of the quadratic function.
- Use real-world examples to illustrate the applications of quadratic functions in space and science.
- Have students work in pairs to compare and contrast different quadratic functions in standard form.